

rently used in creating MS-DOS viruses can be ported to Unix. This includes the creation of automated CAD/CAM virus tools, stealth, polymorphism and armour. The future of viruses on Unix is already hinted at by the wide spread use of Bots and Kill-Bots, (slang term referring to software robots). These programs are able to move from system to system performing their function. Using a Bot as a dropper or creating a virus that includes bot-like capability is simple. With the advent of global networks, the edge between viruses, bots, worms and Trojans will blur. Attacks will be created that use abilities from all of these forms and others to be developed. There have already been cases where people have used audit tools such as COPS and SATAN to attack a system. Combining these tools with a virus CAD/CAM program will allow a fully functional virus factory to create custom viruses and attacks against specific targets such as companies that are disliked by the propitiator. The information services provided by the Internet already provide sufficient information in the form of IP addresses and email domain addresses to identify, locate and attack systems owned by specific entities.

Finally, viruses and worms can provide the perfect format for a hostage shielded denial of service attack. It is well known that an Internet attached system can be made to “disappear” or crash by flooding it with IP packets. Site administrators can protect their systems from crashing by programming their local router to filter out packets from the attacking source. The system will still disappear because legitimate users will be squeezed out by the flood of attack packets, but filtering at the router can at least save the system from crashing. Unfortunately, anyone can masquerade as someone else on the Internet by merely using their IP address. This attack can send a barrage of packets to the target site, each of which has a different source IP address. It is not possible to use a router to filter from this type of attack, but the Internet service provider can trace the source of attack by physical channel without relying upon the IP address. In cooperation with other Internet providers, the attacker can be isolated from the Internet for a